
Problem Set 8

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Problem 1

1.

$$\int_{-2}^1 x^3 \delta(x+1) dx = (-1)^3 = -1.$$

2.

$$\int_{-\infty}^y \delta(x) dx = \begin{cases} 0, & y < 0 \\ 1, & y \geq 0 \end{cases}.$$

3.

$$\int_0^{3\pi/2} \cos x \delta(x - \pi) dx = \cos \pi = -1.$$

4.

$$\int_{\mathbb{R}^3} (r^2 + \vec{r} \cdot \vec{a} + a^2) \delta^{(3)}(\vec{r} - \vec{a}) d\vec{r} = 3a^2.$$

Problem 2

1. Let $u = a(x - x_0)$. Then $du = adx$, so

$$\int_{-\infty}^{\infty} f(x) \delta(a(x - x_0)) dx = \int_{-\infty}^{\infty} \frac{1}{a} f\left(x_0 + \frac{u}{a}\right) \delta(u) du = f\left(x_0 + \frac{0}{a}\right) \frac{1}{a}.$$

Note that the above equations are mild abuse of notation; if $a < 0$, then the bounds in u are the reverse of that in x , and thus shifting them back induces an extra factor of -1 . Thus, if $a < 0$, then the bounds change, inducing an extra negative. We can thus write the more proper solution as

$$\int_{-\infty}^{\infty} f(x) \delta(a(x - x_0)) dx = \frac{f(x_0)}{|a|}.$$

2. We have

$$\int_{-\infty}^{\infty} f(x) \delta(a(x - x_0)) dx = \int_{-\infty}^{\infty} f(x) \frac{1}{|a|} \delta(x - x_0) dx.$$

In order for this equality to hold for all integrable functions f , we must have

$$\frac{1}{|a|} \delta(x - x_0) = \delta(a(x - x_0)).$$

3. We thus have

$$\delta(-x) = \delta(-1(x - 0)) = \frac{1}{|-1|} \delta(x - 0) = \delta(x).$$

Therefore, δ is even.